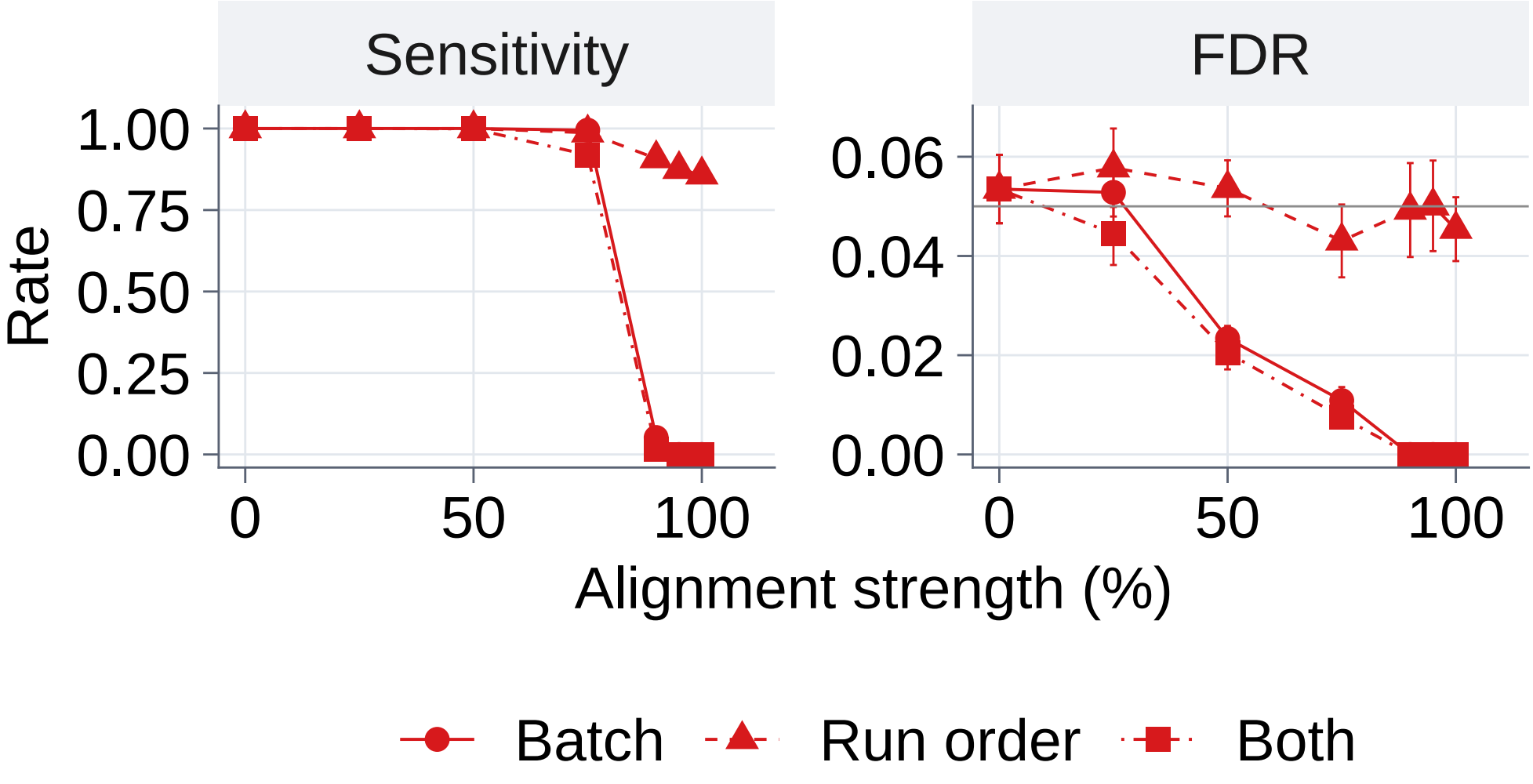F WiNN: detection and false discoveries